Major Domains & Major Landforms of the Earth

Subject: Physical Geography | NCERT Class 6 Ch. 5 & 6 | UPSC & State PSC Core

⚡ 30-Second Snapshot

- Earth's surface functions through the interaction of four spheres: the **Lithosphere** (crust and soil), **Atmosphere** (gaseous envelope), **Hydrosphere** (71% water cover), and the life-bearing **Biosphere**.
- **Africa** is the only continent traversed by the **Tropic of Cancer, Equator, and Tropic of Capricorn**.
- The **Atlantic Ocean** has an **S-shaped basin** with an **indented coastline** that creates natural harbors, making it the most commercially active ocean.
- The **atmosphere** extends to 1,600 km across five layers, dominated by **Nitrogen (78%)** and **Oxygen (21%)**, with both density and temperature falling as altitude rises.
- Landforms are shaped by **internal forces** (uplift and subsidence) and **external forces** (erosion and deposition).
- Mountains are classified as **Fold** (Himalayas, Aravallis), **Block** (Horsts and Graben, like the Rhine Valley), or **Volcanic** (Kilimanjaro, Fujiyama).
- **Plains** (below 200 m elevation) formed by river deposition host the world's densest human populations due to fertile alluvium and level terrain.

🏛 The Four Domains & Ocean Basins

Global Domain / Basin	Structural Architecture	Defining Geographic Markers	Human & Ecological Relevance
Lithosphere	Earth's rocky crust and nutrient-rich topsoil layer.	Divided into 7 continents; highest peak is Mt. Everest (8,848 m).	Supports continental agriculture, mineral extraction, and human habitats.
Hydrosphere	Covers 71% of the surface; 97% is saline ocean water.	Deepest depression is the Mariana Trench (11,022 m) in the Pacific Ocean.	Regulates global climate through continuous waves, tides, and currents.
Atmosphere	Gaseous mantle extending upward to 1,600 km in 5 layers.	Composed of Nitrogen 78%, Oxygen 21%, and trace gases 1%.	Protects the surface from radiation; CO_2 keeps the planet warm via heat absorption.
Biosphere	Narrow contact zone where land, water, and air meet.	Divided broadly into the plant kingdom and the animal kingdom.	The only known planetary zone capable of supporting living organisms.
Atlantic Ocean	S-shaped basin flanked by the Americas, Europe, and Africa.	Irregular, highly indented coastlines create deep natural anchorages.	The most commercially active ocean globally for maritime trade routes.
Arctic Ocean	Positioned within the Arctic Circle around the North Pole.	Connected to the Pacific Ocean through the shallow Bering Strait.	Cold circumpolar marine ecosystem bordered by Eurasia and North America.

📐 Geomorphic Processes & Landform Taxonomy

- **Dual Geomorphic Mechanisms:**
  - **Endogenic (Internal) Forces:** Subterranean tectonic forces that cause vertical and horizontal displacement, uplifting plateaus and sinking rift basins.
  - **Exogenic (External) Forces:** Continuous sculpturing of the surface through **erosion**

(denudation) and **deposition** (aggradation) by water, wind, and glaciers.

- **Taxonomy of Mountain Systems:**
  - **Fold Mountains:** Formed by crustal compression. Young fold mountains have sharp, rugged peaks (Himalayas, Alps); old fold mountains are worn down by long-term weathering (Aravallis, Urals, Appalachians).
  - **Block Mountains:** Formed when crustal fractures cause vertical fault displacement. The uplifted crustal blocks are **Horsts**; the sunken rift blocks are **Graben** (e.g., Vosges Mountains and the Rhine Valley in Europe).
  - **Volcanic Mountains:** Built from erupted lava, volcanic ash, and pyroclastic debris (e.g., Mt. Kilimanjaro in Africa, Mt. Fujiyama in Japan).
- **Plateaus vs. Depositional Plains:**
  - **Plateaus:** Elevated, flat-topped tablelands with steep scarps. High mineral reserves (Chhotanagpur in India; gold/diamonds in Africa). Can host waterfalls (Hundru, Jog) and fertile black lava soils. The Tibetan Plateau is the world's highest (4,000–6,000 m).
  - **Plains:** Low-relief flatlands generally under 200 m above sea level. Built by rivers depositing sand, silt, and clay (alluvium) along their valleys. Ideal for farming and transport networks, making them the most densely populated regions on Earth (e.g., Indo-Gangetic Plains).

⚠ Common UPSC Exam Traps

- **Trap 1 (Continental Latitudinal Crossings):** Africa is the only continent crossed by the Tropic of Cancer, Equator, and Tropic of Capricorn. Asia is crossed by the Equator, Tropic of Cancer, and Arctic Circle, but not the Tropic of Capricorn.
- **Trap 2 (Old vs. Young Fold Mountains):** The Aravalli Range in India and the Appalachians in North America are old, eroded fold mountains with rounded peaks, not block mountains.
- **Trap 3 (Horst and Graben Architecture):** In block faulting, the uplifted block is the Horst, while the down-dropped graben forms the sunken rift valley floor.
- **Trap 4 (Freshwater Availability):** Water covers 71% of Earth, but over 97% of it is saline ocean water. Less than 3% is freshwater, and most of that is locked in polar ice caps or deep underground, leaving only a tiny fraction accessible in rivers and lakes.

🎯 High-Yield Facts Box

Parameter / Feature	Scientific / Geographic Benchmark	Global Significance
Highest Elevation	Mt. Everest (8,848 m)	Tallest subaerial peak, located in the young fold Himalayas.
Deepest Depression	Mariana Trench (11,022 m)	Submarine trench in the Pacific Ocean, deeper than Everest is tall.
Submerged Giant	Mauna Kea, Hawaii (10,205 m)	Volcanic seamount rising from the Pacific floor, taller than Everest from base to peak.
Highest Plateau	Tibetan Plateau (4,000 to 6,000 m)	"Roof of the World"; vast high-altitude tableland in Central Asia.
Indian Polar Outposts	Maitri and Dakshin Gangotri	Permanent scientific research stations in Antarctica.

📝 Prelims Practice Challenge

Q1. Consider the following statements regarding the global distribution of land and water:

1. Africa is the only continent through which the Tropic of Cancer, the Equator, and the Tropic of Capricorn all pass.
2. The Atlantic Ocean features an indented coastline that provides favorable natural conditions for harbors and commercial ports.

3. More than half of the Earth's total freshwater reserve is easily accessible in surface rivers and freshwater lakes.

Which of the statements given above are correct?

- (A) 1 and 2 only
- (B) 2 and 3 only
- (C) 1 and 3 only
- (D) 1, 2, and 3

> **Answer: (A) 1 and 2 only**
> **Explanation:** Statements 1 and 2 are correct because Africa is uniquely traversed by all three primary parallels, and the indented Atlantic coastline supports extensive maritime commerce. Statement 3 is incorrect because over 97% of Earth's water is saline, and of the remaining freshwater, the majority is locked in glaciers, polar ice sheets, or deep underground aquifers.

Q2. With reference to major landforms, consider the following statements:

1. Block mountains are formed when large crustal sections are fractured and vertically displaced, creating horsts and graben.
2. The Aravalli Range in India represents one of the oldest active volcanic mountain systems in the world.
3. Depositional river plains rarely exceed 200 meters above mean sea level and support dense human populations.

Which of the statements given above is/are correct?

- (A) 1 and 2 only
- (B) 1 and 3 only
- (C) 3 only
- (D) 1, 2, and 3

> **Answer: (B) 1 and 3 only**
> **Explanation:** Statements 1 and 3 are correct because vertical fault displacement creates horsts and graben, and low-relief river plains (under 200 m) provide fertile alluvium that supports high population densities. Statement 2 is incorrect because the Aravallis are an ancient, highly eroded fold mountain chain, not a volcanic system.

✏ Mains Practice Question (10 Marks / 150 Words)

Question: *"Explain how internal and external geomorphic processes interact to create diverse landforms on the Earth's crust. Why do depositional river plains support significantly higher population densities than structural plateaus?"*

Structured Answer Framework:

- **Introduction (25–30 words):**
  Define the two primary geomorphic forces: endogenic (internal tectonic uplift and faulting) and exogenic (external weathering, erosion, and deposition by water, ice, and wind).
- **Body Arguments (80–90 words):**
  - *Interaction of Forces:* Endogenic forces build primary landforms by uplifting fold mountains (Himalayas) or vertically displacing crustal blocks (Rhine graben). Exogenic forces immediately begin denuding these uplands through water and glacial erosion, depositing the eroded sediment in lowlands to form flat depositional plains.
  - *Plateau Constraints:* Plateaus are elevated tablelands with steep, rugged escarpments. While rich in mineral wealth, their rocky terrain makes canal construction, transport infrastructure, and mechanized farming challenging.
  - *Plains Advantages:* Depositional plains (under 200 m elevation) provide flat land, deep fertile alluvial soils, perennial river water, and easy conditions for laying road and rail grids. This supports high agricultural yields and industrial corridors, sustaining dense populations like those in the Indo-Gangetic Plains.
- **Conclusion / Way Forward (25–30 words):**

While plateaus supply the raw minerals for heavy industry, fertile river plains provide the food security and transport connectivity that make them the primary population centers of the world.

✍ Notes & Updates